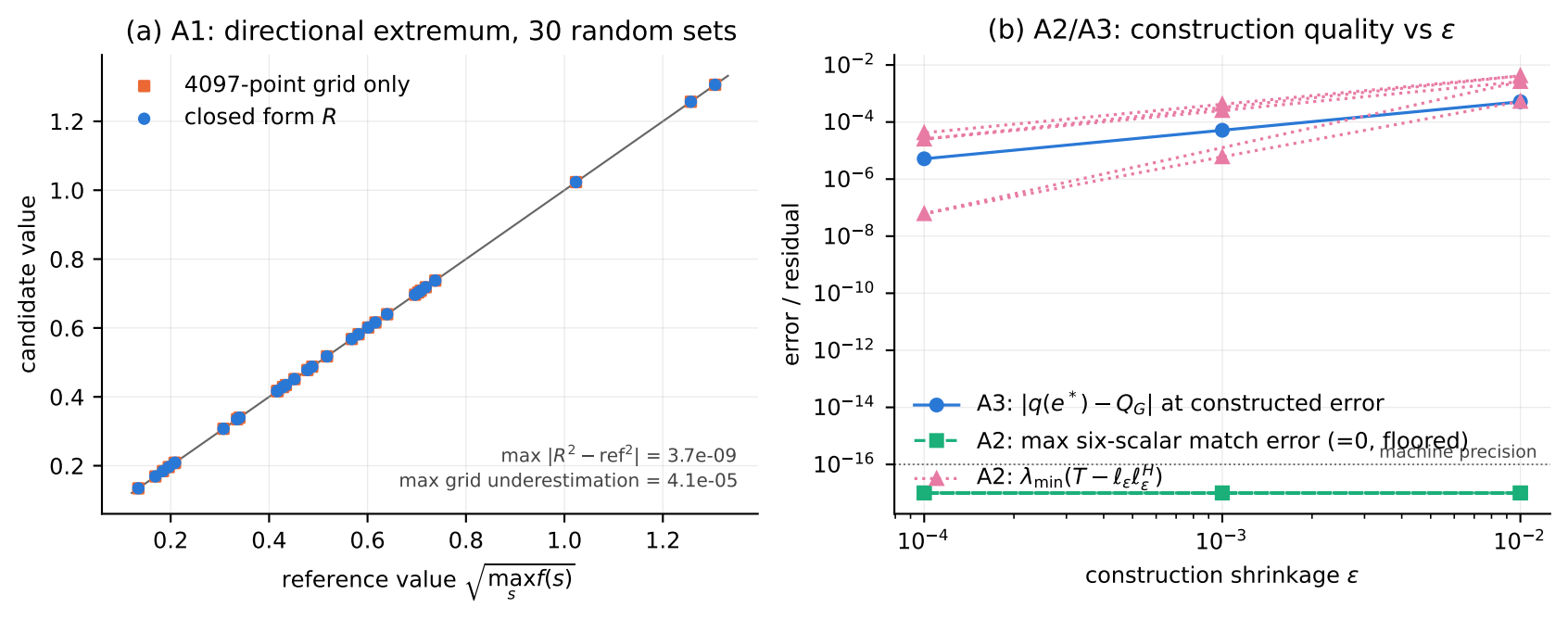
(a) A1: directional extremum, 30 random sets
4097-point grid only
closed form R
max $|R^2 - \mathrm{ref}^2|$ = 3.7e-09
max grid underestimation = 4.1e-05
candidate value
reference value $\sqrt{\max_s f(s)}$
0.2
0.4
0.6
0.8
1.0
1.2
(b) A2/A3: construction quality vs ε
error / residual
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{-16}
A3: $|q(e^*) - Q_G|$ at constructed error
A2: max six-scalar match error (=0, floored)
machine precision
A2: $\lambda_{\min}(T - \ell_\varepsilon \ell_\varepsilon^H)$
10^{-4}
10^{-3}
10^{-2}
construction shrinkage ε